Supplementary methods

A configurable generative transcriptomics framework for spaceflown mice

Jason Trinh

This document records data roles, model configurations, evaluation gates, statistical safeguards, and supporting results.

S1. OSDR discovery and expression ingestion

OSDR data came from the Biological Data API. Eligible profiles were *Mus musculus* bulk RNA-seq with a resolvable flight or ground-control label, processed RSEM expected counts, and tissue or material metadata sufficient for canonicalization.

The API returned 1,631 rows; aggregation of 21 technical replicates left 1,610 biological profiles (835 flight and 775 ground control) from 75 accessions and 24 canonical material classes.

S2. ARCHS4 cohort audit

The source file, `assets/archs4/mouse_gene_v2.5.h5`, contained 997,515 profiles and 53,511 genes. Three reference cohorts were eligible:

Cohort	Profiles	GEO series	Intended use
Control only	23,614	3,213	Conservative sensitivity cohort
Healthy preferred	62,299	5,307	Primary pretraining source
Broad	134,250	15,111	Diversity sensitivity cohort

Nine GEO series linked to eligible OSDR accessions were excluded before selection. This removed 108 ARCHS4 profiles, including all 53 selected profiles from GSE152382 (OSD-457). The paper-parity run selected 17,244 healthy-preferred profiles across 20 tissues. Complete GEO-series grouping assigned 10,150 profiles to training, 2,466 to validation, and 4,628 to test without series overlap.

S3. Landmark panel and normalization

Full-transcriptome TPM was calculated with GENCODE vM39 mouse gene lengths. Landmark selection occurred after TPM calculation and used a deterministic 974-gene mouse mapping of the human L1000 panel. Training-set MaxAbs scaling was applied after landmark selection.

S4. ARCHS4 DDIM configuration

The full configuration is `configs/rna_diffusion/archs4_mouse_paper_parity_osdr_disjoint.yaml`. Architecture and optimization matched the paper-parity implementation; only reference exclusion and grouped splitting changed.

Component	Value
Input genes	974
Hidden layers	8,192; 8,192
Parameters	227,109,786
Dropout	0.1
Diffusion steps	1,000
Beta schedule	Quadratic, 0.0001 to 0.02
Objective	Summed noise MSE
Optimizer	Adam
Learning rate	0.0004783833151836702
Scheduler	OneCycle
Batch size	2,048
Epochs / optimizer steps	15,000 / 75,000
AMP	Enabled
EMA	0.999
Device	NVIDIA A100-SXM4-40GB
Runtime	6,083 seconds
Peak allocated GPU memory	5.93 GB

S5. Factorized OSDR adaptation

The accepted configuration, `configs/rna_diffusion/osdr_factorized_study_lora512_correlation_refine_osdr_disjoint.yaml`, used the Section S4 backbone. A base adapter received 12,000 domain and 4,000 condition steps before the correlation-refinement stage below. All-tissue and muscle-group screens used the refined adapter.

Component	Value
Backbone	Completed ARCHS4 paper-parity DDIM
Conditioning	Tissue, FLT/GC, accession, material type
Domain adapter	LoRA rank 512, alpha 512
Domain stage	4,000 steps, LR 0.00002
Condition stage	1,000 steps, LR 0.00002
Batch size	512
Condition dropout	0.15
Correlation regularization	Weight 10, 256 genes, timestep <= 200
Sampling	100 DDIM steps for evaluation

The split contained 781 training, 536 validation, and 293 locked-test profiles. Every test accession appeared in training, so results measure within-study interpolation. The calibrator used train-only global alignment, hierarchically shrunk accession and tissue means, positive missing-covariance residual noise, no condition-specific fit of means or covariance, and zero clipping for downstream expression.

S6. Distribution and condition gates

Four generation seeds (5020-5023) were declared before opening the locked test. Metrics were gated independently.

Metric	Gate
Gene-correlation agreement	Paper target >= 0.98; finite-sample real-bootstrap floor also reported
Precision	>= 0.95
Recall	>= 0.85
F1	>= 0.90
Adversarial accuracy	0.40 to 0.60
FD / real-split P95	<= 1.0
Pooled effect recovery	Correlation >= 0.30 and direction >= 0.55
Muscle accession recovery	Correlation >= 0.30 and direction >= 0.55
Memorization	Generated fraction below train LOO P01 <= 0.05

Exact repeats are in `source_data/table_s2_locked_ddim_repeats.tsv`. Locked-test means were correlation 0.9744, precision 0.9974, recall 0.9957, F1 0.9966, adversarial accuracy 0.4753, and FD/real-P95 0.0740. All repeats exceeded the finite-sample correlation floor of 0.9497, although the mean missed the paper target of 0.98. Pooled FLT/GC recovery passed in three of four repeats and muscle accession recovery in all four. The preceding validation screen missed its stricter correlation and muscle gates, so broad biological screens remain developmental. Adversarial accuracy denotes an external nearest-neighbor real-versus-synthetic classifier, not the WGAN critic; 0.5 indicates chance separation.

S7. Configurable benchmark and comparator models

The 463-row staged planner varied inputs, normalization, transformation, scaling, genes, harmonization, training data, accession and tissue scope, FLT/GC and study conditioning, balancing, seed, and synthetic ratio. Only branches passing lower-cost screens received full training.

Liver harmonization benchmark

Nine arms used the same 974-gene DDIM, seed, A100 device, and split: 119 training, 50 validation, and 70 unopened locked-test profiles. No arm passed all fidelity, pooled-condition, and accession-effect gates.

Method	Corr.	Precision	Recall	F1	AA	FD/real P95	Fidelity gate	Accession gate
No harmonization, TPM	0.283	0.200	0.960	0.331	0.850	2.052	Fail	Fail
Ilangovan study z-score	0.278	0.160	1.000	0.276	0.770	0.716	Fail	Fail
Mentor two-stage z-score	0.348	0.440	1.000	0.611	0.690	0.977	Fail	Fail
ComBat	0.004	0.040	1.000	0.077	0.810	63.185	Fail	Fail
ComBat-seq	0.067	0.020	1.000	0.039	0.850	156.647	Fail	Fail
MBatch Median Polish	0.009	0.020	1.000	0.039	0.930	205.470	Fail	Fail
MBatch Empirical Bayes	0.001	0.000	1.000	0.000	0.850	60.625	Fail	Fail
MBatch ANOVA	-0.003	0.020	1.000	0.039	0.870	44.716	Fail	Fail
MOBER	0.808	0.260	1.000	0.413	0.770	33.311	Fail	Fail

ComBat, ComBat-seq, and MBatch were transductive; MOBER was inductive. None passed the joint criteria. Tables S13-S14 contain full labels and metrics.

WGAN-GP and GeneJEPA screens

The WGAN used the Viñas et al. topology: 64-dimensional noise, two 256-unit generator and critic layers, five critic updates, gradient-penalty weight 10, and batch size 32. RMSProp used learning rate 0.0005, alpha 0.9, and epsilon 1e-7 for at most 2,000 epochs. Released-code early stopping began at epoch 1, ran every five epochs, and allowed ten failed checks. Across six seeds, mean correlation, precision, recall, F1, adversarial accuracy, and FD/real-P95 were 0.9759, 0.9764, 0.9938, 0.9850, 0.6362, and 0.1439. Pooled FLT/GC recovery passed 6/6 repeats; accession-aware muscle recovery passed 0/6. No repeat passed full fidelity, so the locked test remained unopened. Table S15 contains exact repeats.

GeneJEPA used the released 768-dimensional architecture with 512 latents, 24 blocks, 12 heads, 4,096 tokens, mask ratio 0.45, minimum context 512, and minimum target size 16 per block. Its one-epoch mouse screen used 43,744 replacement-sampled exposures, batch size 92 with two-step gradient accumulation, learning rate 0.0001, weight decay 0.0002, cosine scheduling with 5% warmup, bfloat16 AMP, EMA 0.992 to 0.9995 after 2,000 warmup steps, no weighted sampling, and similarity/variance/covariance weights 1/25/1. Held-out tissue balanced accuracy was 0.703 versus 0.839 from expression. It has no expression decoder.

S8. Generated-feature workflow

The funnel tested pooled utility, tissue development, and real-data association. Pooled classifiers used real, generated, or combined profiles. Tissue screens compared `real_only`, `generated_only`, equal-weight `real_plus_generated`, consensus-ranked `guided_real_only`, and `guided_low_weight` with recentered synthetic profiles at 0.05 total weight.

Eight outer splits evaluated real profiles within accession-condition strata; inner loops selected feature count, regularization, and ranking. Eligibility required nonworse balanced accuracy, AUROC, and average precision versus real-only; ties met this rule without a mean gain. Accessions could occur in both partitions, so this tested within-study development rather than study transfer.

Stable genes required selection frequency >= 0.50 and sign agreement >= 0.75. Reinforced genes were stable in real-only and selected synthetic-guided arms with matching directions and a real meta-effect. Synthetic-promoted genes were stable only with eligible guidance and had a real meta-effect. These labels classify selection, not novelty; random-effects and LOO tests followed.

Supplementary Table S9. Complete canonical-tissue utility screen.

Tissue	n (FLT/GC)	Selected arm	BA real/selected	AUROC real/selected	AP real/selected	Status
Adrenal gland	31 (16/15)	Generated only	0.781 / 0.922	0.906 / 0.984	0.918 / 0.988	Eligible improvement
Bone	30 (15/15)	Guided ranking; real fit	0.656 / 0.734	0.797 / 0.828	0.858 / 0.865	Eligible improvement
Bone marrow	20 (10/10)	Generated only	0.969 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible improvement
Brain	45 (22/23)	Generated only	0.688 / 0.740	0.712 / 0.802	0.731 / 0.788	Eligible improvement
Brown adipose tissue	20 (10/10)	Generated only	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible tie
Cecum	16 (8/8)	Real only	0.875 / 0.875	0.938 / 0.938	0.958 / 0.958	Real-only retained
Cerebellum	44 (23/21)	Generated only	0.529 / 0.602	0.621 / 0.696	0.729 / 0.764	Eligible improvement
Colon	45 (23/22)	Real only	0.656 / 0.656	0.677 / 0.677	0.730 / 0.730	Real-only retained
Eye	18 (9/9)	Generated only	0.594 / 0.781	0.562 / 0.781	0.729 / 0.865	Eligible improvement
Heart	42 (21/21)	Generated only	0.594 / 0.719	0.625 / 0.750	0.642 / 0.778	Eligible improvement
Hippocampus	30 (16/14)	Generated only	0.698 / 0.703	0.740 / 0.781	0.849 / 0.864	Eligible improvement
Kidney	111 (56/55)	Guided ranking; 5% synthetic	0.654 / 0.707	0.680 / 0.771	0.683 / 0.799	Eligible improvement
Liver	197 (101/96)	Real only	0.721 / 0.721	0.764 / 0.764	0.773 / 0.773	Real-only retained
Lung	63 (32/31)	Generated only	0.664 / 0.742	0.680 / 0.830	0.684 / 0.833	Eligible improvement
Mammary gland	20 (8/12)	Generated only	0.771 / 0.885	0.854 / 0.958	0.875 / 0.958	Eligible improvement
Optic nerve	29 (19/10)	Guided ranking; 5% synthetic	0.769 / 0.819	0.863 / 0.950	0.950 / 0.983	Eligible improvement
Retina	63 (37/26)	Guided ranking; 5% synthetic	0.587 / 0.723	0.620 / 0.762	0.756 / 0.846	Eligible improvement
Skeletal muscle, pooled	149 (74/75)	Guided ranking; real fit	0.861 / 0.932	0.924 / 0.961	0.908 / 0.945	Eligible improvement
Skin	125 (66/59)	Real + generated	0.671 / 0.756	0.691 / 0.768	0.747 / 0.808	Eligible improvement
Spleen	86 (44/42)	Real + generated	0.517 / 0.648	0.540 / 0.704	0.589 / 0.749	Eligible improvement
Thymus	94 (51/43)	Guided ranking; 5% synthetic	0.733 / 0.844	0.818 / 0.887	0.862 / 0.918	Eligible improvement
White adipose tissue	20 (10/10)	Generated only	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible tie

Supplementary Table S6. Complete anatomical muscle-group utility screen.

Tissue	n (FLT/GC)	Selected arm	BA real/selected	AUROC real/selected	AP real/selected	Status
EDL	24 (12/12)	Real only	0.917 / 0.917	1.000 / 1.000	1.000 / 1.000	Real-only retained
Gastrocnemius	25 (10/15)	Guided ranking; 5% synthetic	0.604 / 0.646	0.646 / 0.750	0.731 / 0.798	Eligible improvement
Quadriceps	35 (18/17)	Real only	0.750 / 0.750	0.880 / 0.880	0.916 / 0.916	Real-only retained
Soleus	41 (22/19)	Real + generated	0.925 / 0.963	0.980 / 0.980	0.980 / 0.986	Eligible improvement
Tibialis anterior	24 (12/12)	Real + generated	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible tie

Sample counts are shown as total development profiles followed by flight/ground-control counts. Small cohorts and ceiling-level scores remain exploratory even when a synthetic arm is eligible.

S9. Random-effects reporting and LOO sensitivity

For gene `g` and accession `a`, the real effect was `delta[g,a] = mean(FLT,g,a) - mean(GC,g,a)`. Accession effects were combined by random-effects meta-analysis. Within each tissue, Benjamini-Hochberg adjustment was applied to 974 gene-level P values; BH FDR < 0.05 was the primary rule. Generated profiles never entered this model.

The inventory annotates reinforced, synthetic-promoted, real-only, synthetic-selected without real-direction support, and unselected associations. It also reports study count, direction fraction, `tau2`, and `I2`; mixed directions qualify but do not negate a significant pooled effect. LOO refitted the model after removing each accession. LOO stability required maximum LOO FDR < 0.05 without sign reversal and was a sensitivity label, not an inclusion rule.

S10. Reactome analysis

The official mouse GMT was generated from `ReactomePathways.txt` and `Ensembl2Reactome_All_Levels.txt`, restricted to *Mus musculus*, `R-MMU-*` pathways, and `ENSMUSG*` genes.

Hypergeometric enrichment used the 974-gene landmark panel as background. Benjamini-Hochberg FDR was applied separately by tissue and selected-gene set. Reactome parent and child terms overlap. Counts of significant rows are therefore not counts of independent biological discoveries.

S11. Skeletal-muscle group analysis

The factorized DDIM and three frozen synthetic development views were reused. No additional neural network was trained for the muscle-group screen.

Group	Profiles	Accessions	FLT	GC
EDL	24	2	12	12
Gastrocnemius	25	3	10	15
Quadriceps	35	4	18	17
Soleus	41	3	22	19
Tibialis anterior	24	2	12	12

Table S6 gives the exact utility results. Soleus reinforced FLT-lower `Bdh1`, `Ech1`, `Bnip3`, and `Decr1` and FLT-higher `Tpm1` across all three accessions. All except `Decr1` passed LOO sensitivity; `Mef2c` and `Pxmp2` were not promoted. Gastrocnemius promoted `Fhl2` and `Nfkbia`; tibialis anterior reinforced `Cdkn1a`, `St3gal5`, and `Bnip3` and promoted `Cebpd`. None passed LOO FDR. EDL and quadriceps retained real-only arms.

LOO refitted only the real-data meta-analysis; it did not remove an accession from generator adaptation. The muscle findings therefore remain developmental pending a fully excluded study.

S12. Random-effects BH-FDR gene inventories

Table S11 contains all real-data random-effects associations with BH FDR < 0.05; Table S12 gives counts for every analysis unit. The 459 tissue-gene rows comprise 202 associations across 10 of 22 canonical tissues and 257 across five muscle groups. They are not independent discoveries because genes and samples can recur across analyses. Direction was unanimous for 363 rows and mixed for 96; this was an annotation, not a filter.

Analysis unit	BH-FDR genes	FLT higher	FLT lower	Unanimous direction	Synthetic-promoted	Reinforced
Adrenal gland	22	1	21	21	1	1
Eye	8	4	4	8	0	1
Heart	4	2	2	4	0	0
Kidney	3	3	0	1	1	1
Liver	19	8	11	0	0	0
Retina	5	5	0	5	0	0
Skeletal muscle, pooled	26	12	14	5	4	8
Skin	3	2	1	1	1	0
Spleen	34	32	2	21	3	1
Thymus	78	37	41	57	13	3
EDL	136	14	122	130	0	0
Gastrocnemius	15	8	7	13	2	0
Quadriceps	29	25	4	22	0	0
Soleus	29	5	24	27	0	5
Tibialis anterior	48	42	6	48	1	3

Table S10 is the 49-row synthetic-informed subset: 26 promoted and 23 reinforced results with supporting real effects. Attribution was suppressed when the generated arm failed eligibility. The remaining inventory contains 34 real-only selections, 370 BH-significant unselected results, and six synthetic selections without real-direction support. Below, direction is the real random-effects estimate; an asterisk marks disagreement among accessions.

Synthetic-promoted genes

Analysis unit	FLT higher	FLT lower
Adrenal gland	None	`Psmb8`
Kidney	`Inpp4b`*	None
Skeletal muscle, pooled	None	`Klhl21`*, `Mapkapk5`*, `Reep5`*, `Itgb5`*
Skin	`Plscr1`	None
Spleen	`Rai14`, `Myl9`*, `Ptprk`	None
Thymus	`Hsd17b11`*, `Etv1`	`Nusap1`, `Stmn1`, `Birc5`, `Cdk1`, `Top2a`, `Ccnb2`, `Aurka`, `Ccne2`, `Kif20a`, `Pcna`, `Ccnf`
Gastrocnemius	`Nfkbia`	`Fhl2`
Tibialis anterior	`Cebpd`	None

Reinforced genes

Analysis unit	FLT higher	FLT lower
Adrenal gland	None	`Tspan4`
Eye	None	`Klhl21`
Kidney	`Slc37a4`	None
Skeletal muscle, pooled	`Sox4`, `Cebpd`*, `Sh3bp5`, `Prkcd`, `Arid5b`*, `Sesn1`*, `Tle1`*	`Bphl`*
Spleen	`Loxl1`	None
Thymus	`Snx7`	`Ube2c`, `Gmnn`
Soleus	`Tpm1`	`Bdh1`, `Ech1`, `Bnip3`, `Decr1`
Tibialis anterior	`Cdkn1a`, `St3gal5`, `Bnip3`	None

Heart, liver, retina, EDL, and quadriceps had BH-FDR genes but none was synthetic-informed. Bone, bone marrow, brain, brown adipose tissue, cecum, cerebellum, colon, hippocampus, lung, mammary gland, optic nerve, and white adipose tissue had no BH-FDR gene in the landmark panel. Tables S11-S12 retain the complete results, including pooled muscle, liver's 19 real-only associations, skin's promoted `Plscr1`, and the null lung inventory.

Machine-readable Tables S1-S15 are under `source_data/`.

S13. Supplementary figures

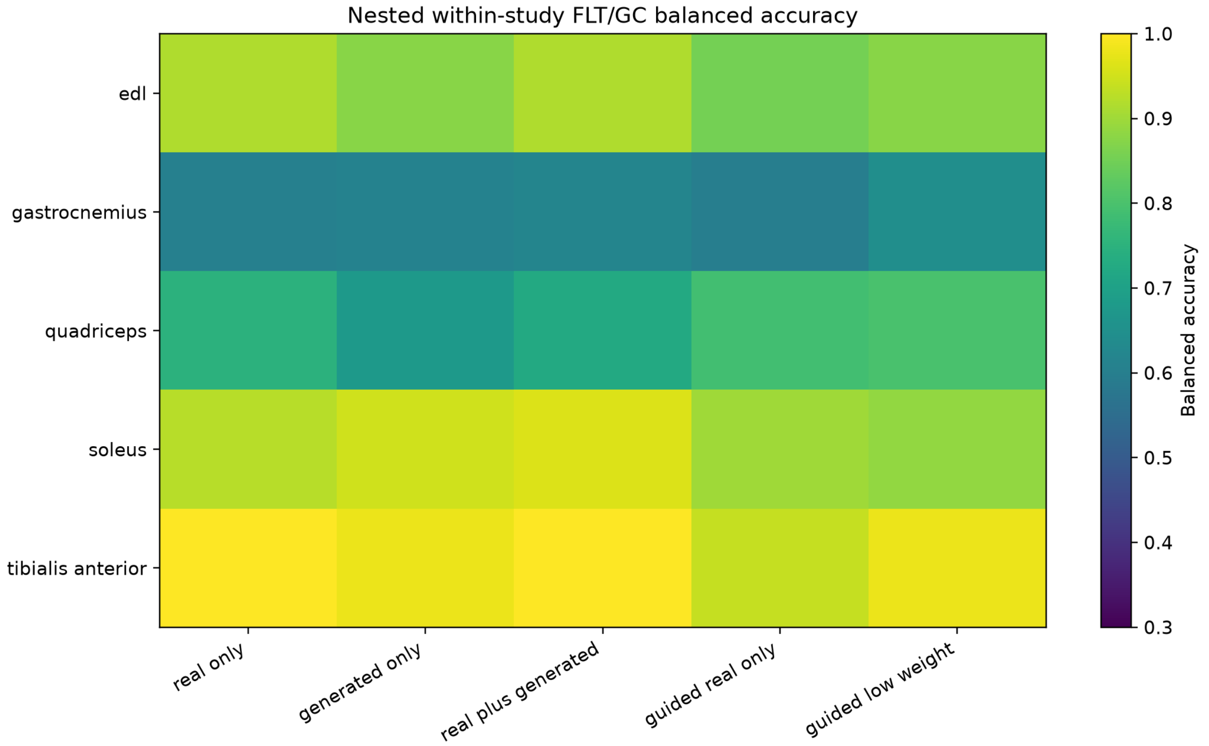

Figure S1. Repeated nested muscle-group balanced accuracy. Each row is a muscle group and each column is a downstream use of real or generated profiles. Arm selection also required nonworse AUROC and average precision.

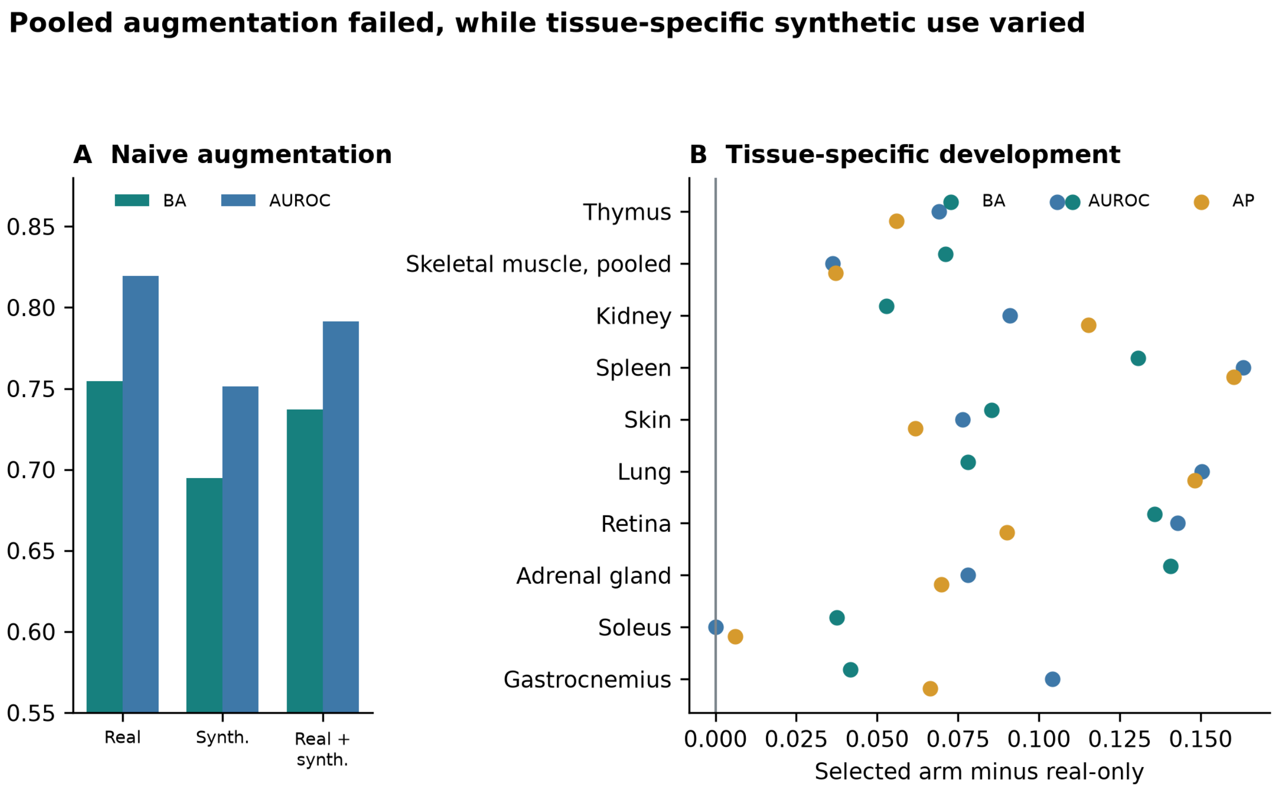

Figure S2. Downstream utility of generated expression. (A) Direct pooled augmentation on the locked real-profile test. (B) Selected-arm changes in balanced accuracy, AUROC, and average precision across repeated tissue-specific development splits. All evaluations used real profiles.